An Internship Report

Of

Siemens Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410030022

Name of Student: REVAT BORBORUAH

Batch: 2024-2029

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CANDIDATE'S DECLARATION

I, REVAT BORBORUAH, do hereby declare that the internship report titled "Siemens Data Science Master Virtual Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-
Student Name:
REVAT BORBORUAH
Roll No.: 2410030022

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and Siemens for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Data Science using the RapidMiner AI Hub. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-
Student Name:
REVAT BORBORUAH
Roll No.: 2410030022

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3. INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF EDUCATION
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अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education

NATIONAL INTERNSHIP PORTAL

EduSkills®
Nation Building Through Skills

AICTE - EDUSKILLS 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Revat Borboruah

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

G.V. Pai

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

AICTE - EDUSKILLS VIRTUAL INTERNSHIP

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of "Siemens Data Science Master", supported by Siemens. The internship was carried out from June 2026 to August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of this internship was to build a strong practical foundation in data science and applied machine learning using the RapidMiner AI Hub platform – covering the identification of business problems suitable for AI and ML, structured data engineering, foundational and advanced machine learning techniques, and enterprise-level deployment, monitoring, and administration of an AI platform.

The internship followed a structured, week-wise curriculum combining conceptual learning with hands-on assignments, weekly assessments, and project documentation, culminating in a Final Credential Validation / Internship Grade Point Assessment. On successful completion of all the requirements, a Virtual Internship Certificate with an "Outstanding" grade was awarded by EduSkills Academy under the patronage of AICTE and the National Internship Portal.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme "Nation Building Through Skills". EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the Data Science Master track, EduSkills collaborates with Siemens Industry Software to design an industry-aligned curriculum built around the RapidMiner AI Hub – Siemens' enterprise data science and AI platform – that helps engineering students build practical, end-to-end data science capability, from business use-case discovery through to enterprise-grade model deployment and platform administration.

4.3 Problem Statement

Although a large number of engineering students learn the theoretical foundations of statistics and machine learning during their academic coursework, very few get structured, hands-on exposure to the complete enterprise data science lifecycle – identifying business problems suitable for AI and ML, engineering and preparing data at scale, building and validating machine learning models, and deploying, monitoring, and administering those models on a production-grade platform. The problem addressed by this internship was to practically learn and apply this end-to-end workflow using an industry-standard enterprise platform, RapidMiner AI Hub, rather than understanding it only at a theoretical level.

4.4 Project Objectives

- To learn how to identify business problems suitable for AI and machine learning solutions.
- To learn how to access, load, transform, and calculate data as part of a structured data engineering workflow.
- To build foundational machine learning skills, including model building and evaluation.
- To master advanced data handling and automation techniques for large and complex datasets.
- To learn advanced machine learning techniques for improved model performance.

- To learn how to deploy and monitor machine learning models in a production-like environment.
- To learn how to configure and manage the RapidMiner AI Hub platform.
- To gain exposure to enterprise-level platform administration operations.
- To successfully clear the Final Credential Validation / Internship Grade Point Assessment.

4.5 Scope of the Project

The scope of this internship spans the complete RapidMiner-based data science workflow – starting from identifying business problems suitable for AI and ML, moving through structured data engineering (accessing, loading, transforming, and calculating data), and extending into foundational and advanced machine learning model building. It further covers mastering advanced data handling and automation, deploying and monitoring machine learning models, and configuring and administering the RapidMiner AI Hub at an enterprise level. The internship was limited to a virtual, simulation/lab-based learning environment provided by Siemens and EduSkills Academy, and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Data Science Platform: RapidMiner Studio, RapidMiner AI Hub
- Data Engineering: Data access, loading, transformation, and calculation pipelines
- Machine Learning: Supervised and unsupervised learning techniques, model building, validation, and tuning
- Deployment & Monitoring: Model deployment workflows and monitoring dashboards on RapidMiner AI Hub
- Platform Administration: Enterprise configuration and administration of RapidMiner AI Hub
- Learning Platform: AICTE – EduSkills Virtual Internship Portal, weekly assessments, and project documentation

4.7 System Architecture

The general architecture followed during the internship for the Siemens Data Science Master learning track can be summarized in the following layers:

- Business Understanding Layer: Identifying business problems and use cases suitable for AI and machine learning solutions.
- Data Layer: Accessing, loading, transforming, and calculating data as part of a structured data engineering pipeline.
- Modeling Layer: Building foundational and advanced machine learning models to address the identified business problem.
- Automation Layer: Mastering advanced data handling and automation to scale data preparation.
- Deployment Layer: Deploying trained models and monitoring their performance in a production-like setting.
- Platform Layer: Configuring and administering the RapidMiner AI Hub at an enterprise level.
- Assessment Layer: Weekly assessments and project documentation culminating in a Final Credential Validation / Internship Grade Point Assessment.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, and assessments, culminating in a Final Credential Validation, as summarized in the table below.

Week	Module	Description
1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
3	Machine Learning Professional	Build foundational machine learning skills.
4	Data Preparation Master	Master advanced data handling and automation.
5	Machine Learning Master	Learn advanced machine learning techniques.
6	Applications & Use Cases Master	Deploy and monitor ML models.
7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
8	Platform Administration Master	Enterprise-level platform operations.

Final Credential Validation / Internship Grade Point Assessment

Note: There was no stipend associated with this internship.

Each week concluded with assessments and project documentation to evaluate conceptual and practical understanding. In the final phase, a Final Credential Validation / Internship Grade Point Assessment was conducted, and the overall grade was based on performance across the weekly modules and this final assessment.

4.9 Expected Outcomes

- Ability to identify business problems and use cases suitable for AI and machine learning solutions.
- Practical understanding of data engineering, including accessing, loading, transforming, and calculating data.
- Capability to build, evaluate, and improve foundational and advanced machine learning models.
- Hands-on exposure to mastering advanced data handling and automation for large datasets.
- Skill to deploy and monitor machine learning models in a production-like environment.
- Ability to configure and administer the RapidMiner AI Hub at an enterprise level.
- A completed set of weekly learning modules, along with successful clearance of the Final Credential Validation / Internship Grade Point Assessment, resulting in an “Outstanding” grade.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Siemens Data Science Master, are attached in Chapter 3 of this report and below as further proof of participation and completion.

5. BIBLIOGRAPHY/REFERENCES

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